XAI Course Project Proposal

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1. Project Overview

We propose implementing a **stability-enhanced LIME framework inspired by MINDFUL-LIME** to explain predictions of a **RuBERT-based transformer model** in **Russian-language sentiment classification**.

The project will include:

- Original implementation of Vanilla LIME for text
- Stability-enhanced modification inspired by **MINDFUL-LIME**
- Application to RuBERT for 3-class sentiment classification
- Comparative analysis with optional RuRoBERTa (if resources allow)
- Visual word-level explanations
- Quantitative evaluation of explanation stability and faithfulness
- Public GitHub repository with reproducible code

Our work is grounded in:

- **Ribeiro et al. (2016)** — original LIME framework
- **MINDFUL-LIME** — stability and localization improvements
- **ModernBERT-XAI** — integration of transformer models with LIME interpretability

We extend these ideas to the Russian language domain.

2. Components

2.1 Application Domain

Domain: Natural Language Processing — Sentiment Analysis

Task: 3-class sentiment classification (positive / negative / neutral)

Dataset: *Russian Sentiment Analysis Dataset (Kaggle)*

- Balanced across three classes
- Contains labeled Russian-language reviews and comments
- Columns: `text`, `label`, `src`

Motivation: Although sentiment analysis is widely studied in English, fewer works explore explanation stability for transformer-based models in morphologically rich languages such as Russian. Interpretability is essential in applications such as product review analysis, public opinion monitoring, and content moderation.

Balanced class distribution ensures unbiased model training and reliable explanation comparison across sentiment categories.

2.2 Machine Learning Model

Model Type: Transformer-based text classifier

Primary Model:

- **RuBERT (DeepPavlov/rubert-base-cased)**
- Fine-tuned for 3-class classification

Optional Model (if resources allow):

- RuRoBERTa for comparative analysis

Why RuBERT:

- Pretrained on large Russian corpora
- Strong baseline for Russian NLP
- Compatible with HuggingFace ecosystem
- Allows investigation of subword-token interpretability challenges

Need for XAI: Transformer models operate as high-dimensional black boxes. WordPiece tokenization and contextual embeddings make model decisions difficult to interpret. Understanding which words drive predictions is crucial for trust and bias detection.

2.3 Proposed XAI Method

Baseline Method: LIME (Ribeiro et al., 2016)

LIME explains individual predictions by approximating the model locally with an interpretable surrogate model.

For text:

- Perturb input by removing words
- Generate neighborhood samples
- Fit sparse linear surrogate model
- Extract word importance scores

LIME Objective Function

$$\xi(x) = \underset{g \in G}{\operatorname{argmin}} L(f, g, \pi_x) + \Omega(g)$$

Where:

- f — black-box model (RuBERT)
- g — interpretable surrogate model
- π_x — locality weighting function
- $\Omega(g)$ — complexity regularization term

Stability-Enhanced LIME (Inspired by MINDFUL-LIME)

Instead of purely random perturbations used in standard LIME, we introduce structure-aware perturbations inspired by MINDFUL-LIME. In text, structure is defined via token adjacency or syntactic dependency relations.

MINDFUL-LIME extends the original LIME framework by addressing one of its key limitations: instability and poor localization precision of explanations. In the original paper, the authors focus on medical image analysis and introduce a structure-aware perturbation strategy based on superpixel graphs. Instead of treating superpixels as independent and randomly sampled features, MINDFUL-LIME models spatial relationships between superpixels using graph structures. This allows the method to generate more coherent and anatomically meaningful perturbations, leading to improved explanation stability and localization accuracy.

The key insight of MINDFUL-LIME is that perturbations should respect the inherent structure of the input domain rather than relying on purely random feature removal.

We adapt this structural perturbation principle to the text domain.

In sentiment analysis, words are not independent features; they are connected through syntactic, semantic, and positional relationships. Inspired by the superpixel graph formulation in MINDFUL-LIME, we propose structure-aware perturbations for text by introducing controlled masking strategies that respect linguistic structure.

Specifically, instead of randomly removing individual words as in standard LIME, we introduce:

- Phrase-level perturbations based on adjacent word grouping
- Local neighborhood constraints to preserve contextual coherence

In this setting, the "graph" analogue is defined over textual units (words or phrases), where edges represent structural relationships such as adjacency or syntactic dependency.

Relation to ModernBERT-XAI

The *ModernBERT-XAI* paper proposes a synergistic approach combining:

- Layer-wise learning

- SHAP-LIME interpretability

We reference this work to contextualize:

- Transformer-specific explanation challenges
- Hybrid interpretability strategies
- Model-aware explanation refinement

Unlike ModernBERT-XAI, our focus is:

- Stability analysis
- Model-agnostic enhancement of LIME
- Application to Russian transformers

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3. Deliverables

- Fine-tuned RuBERT sentiment classifier
- Original implementation of LIME for Russian text
- Stability-enhanced LIME module
- 3–5 qualitative explanation case studies
- Quantitative validation using:
  - **Stability score** (variance across runs)
  - **Faithfulness metric** (confidence drop after removing top-k words)
  - Explanation sparsity comparison
- Optional RuRoBERTa comparison
- Public GitHub repository

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4. Timeline

Feb 12: Proposal submission

March 12:

- Dataset preprocessing
- RuBERT fine-tuning
- Baseline LIME implementation

April 1:

- Stability-enhanced LIME implementation
- Initial evaluation

April 15:

- Quantitative analysis
- Visualization and comparison
- Optional RuRoBERTa experiments

April 21: Final presentation

Project Rationale Summary

This project:

- Extends LIME to Russian-language transformers
- Adapts MINDFUL-LIME stability principles to NLP
- Positions itself relative to ModernBERT-XAI
- Provides both qualitative and quantitative evaluation
- Addresses explanation reliability, not just visualization